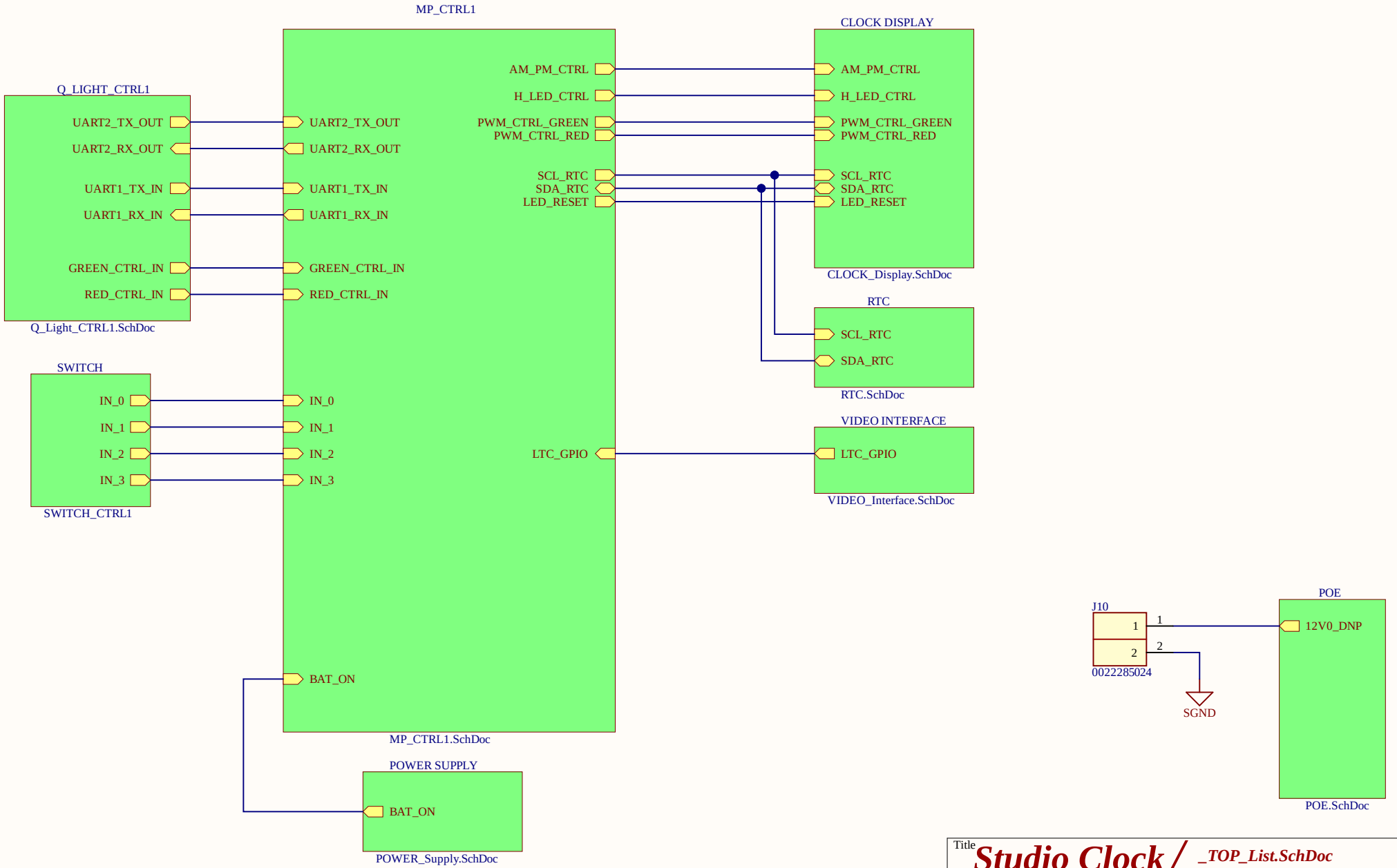
TOP LIST



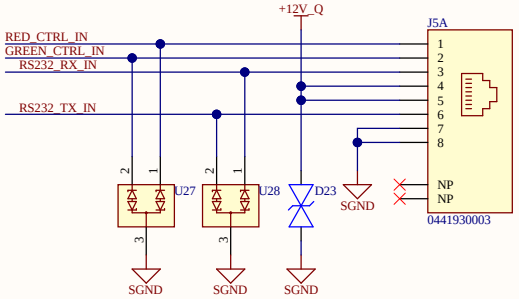
_Top_list.SchDoc

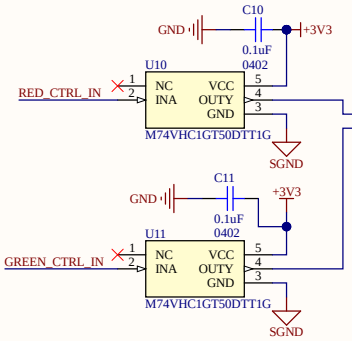
Title	**Studio Clock /** *Title.SchDoc*		
Size A4	Number		Revision *V1.00*
Date:	2.13.2024	Sheet of	*1 / 16*
File:	D:\Dropbox\..\Title.SchDoc	Drawn By:	

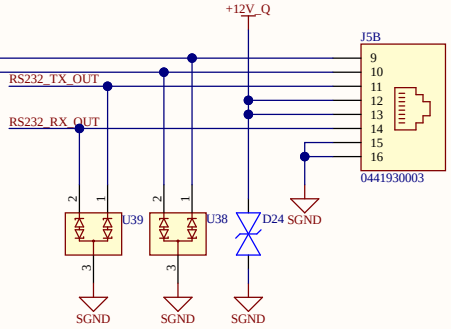


Title	**Studio Clock / _TOP_List.SchDoc**	
Size: A4	Number:	Revision: **V1.00**
Date: 2.13.2024	Sheet of **2 / 16**	
File: D:\Dropbox\.._TOP_List.SchDoc	Drawn By:	

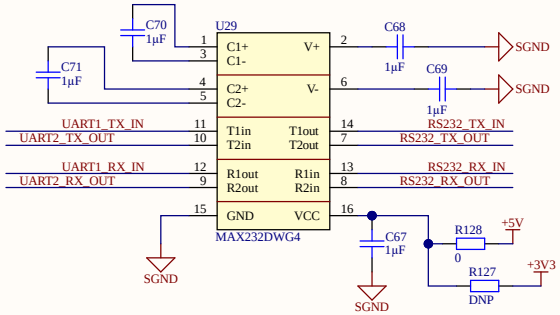
RX & TX are REVERCED!!!





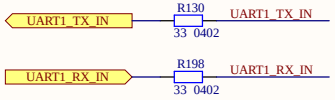


Qlight
Dual RJ45 Connector

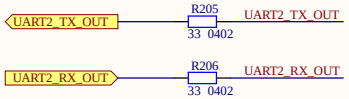


RS232

Input data



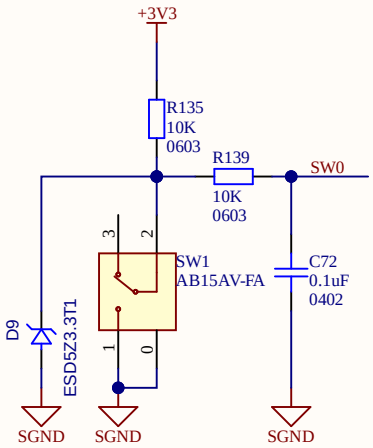
Output data

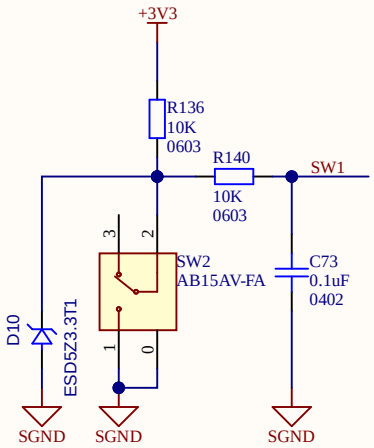


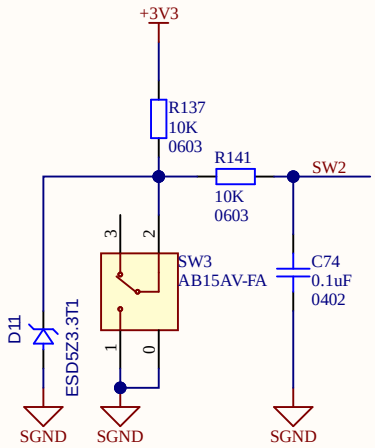


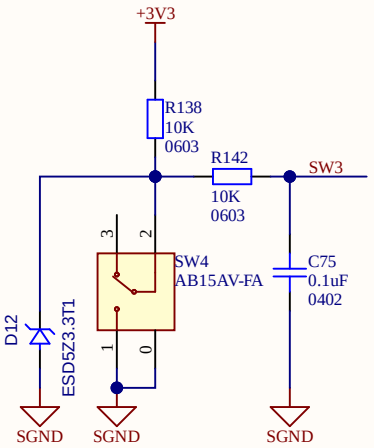
Title: Studio Clock / Q_Light_CTRL1.SchDoc		
Size: A3	Number:	Revision: V1.00
Date: 2.13.2024	Sheet of: 3 / 16	
File: D:\Dropbox\..\Q_Light_CTRL1.SchDoc	Drawn By:	

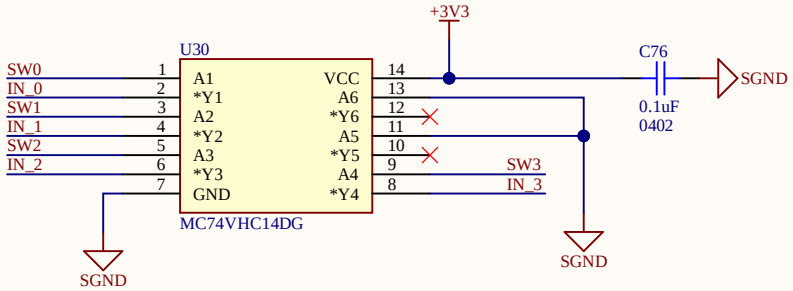
Switches

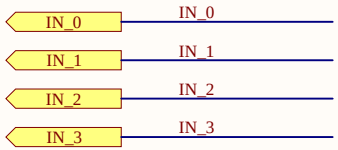




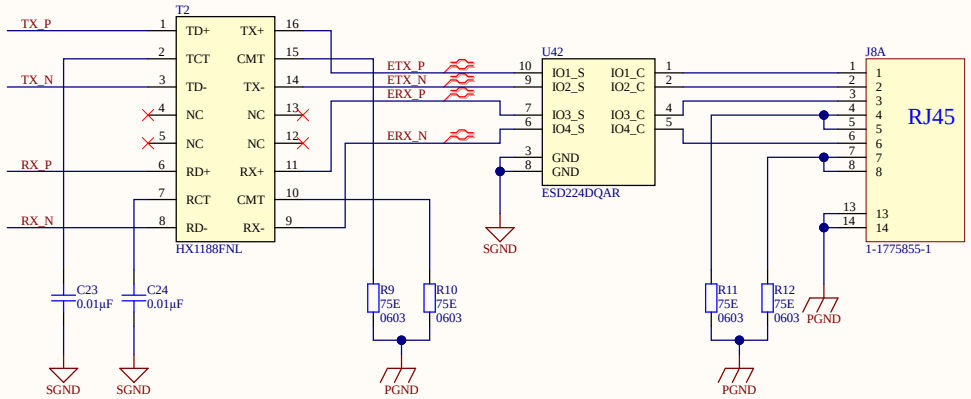


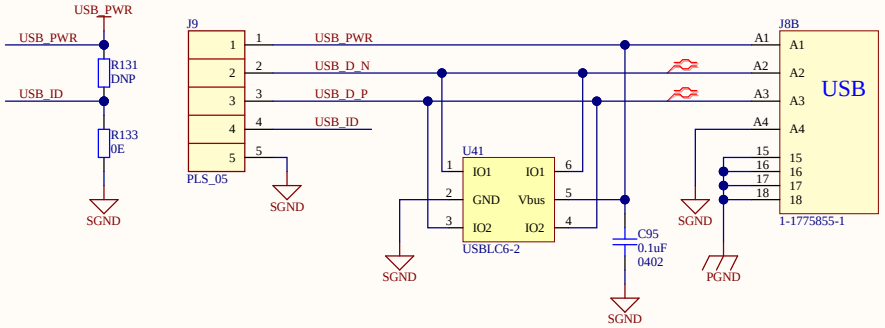




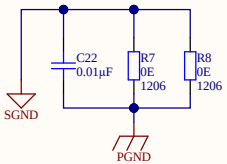


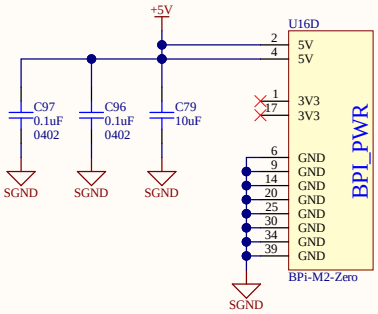
Title	Studio Clock / SWITCH_CTRL1.SchDoc		
Size A4	Number	Revision	V1.00
Date: 2.13.2024		Sheet of	4 / 16
File: D:\Dropbox\..\SWITCH_CTRL1.SchDoc		Drawn By:	

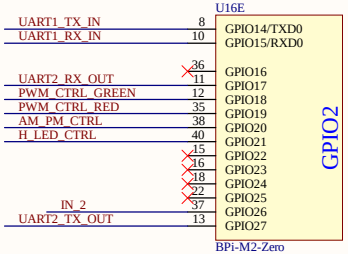




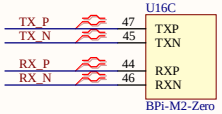
RJ45+USB CONNECTOR

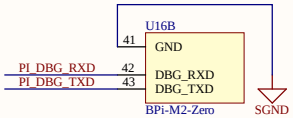


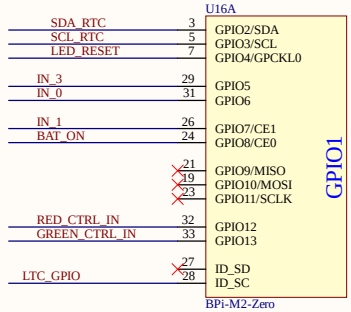




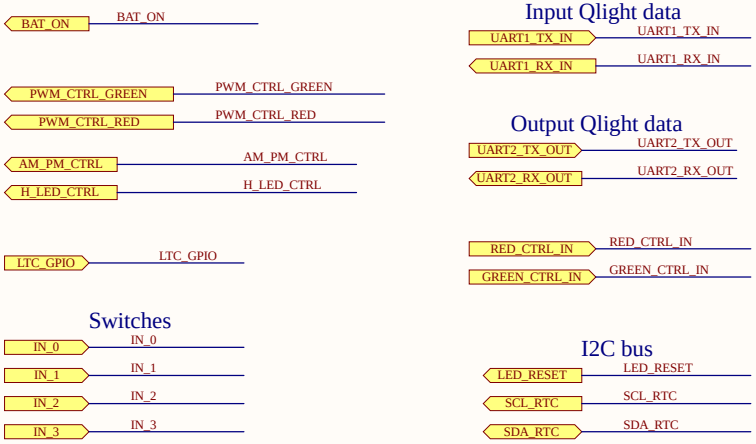
ADD UART2(TX-13pin/RX-11pin) - to QlightOUT connector



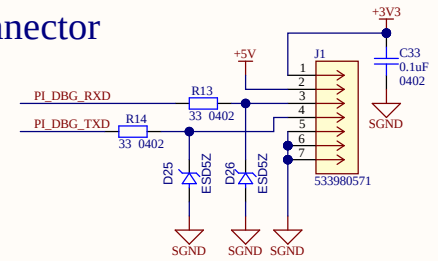




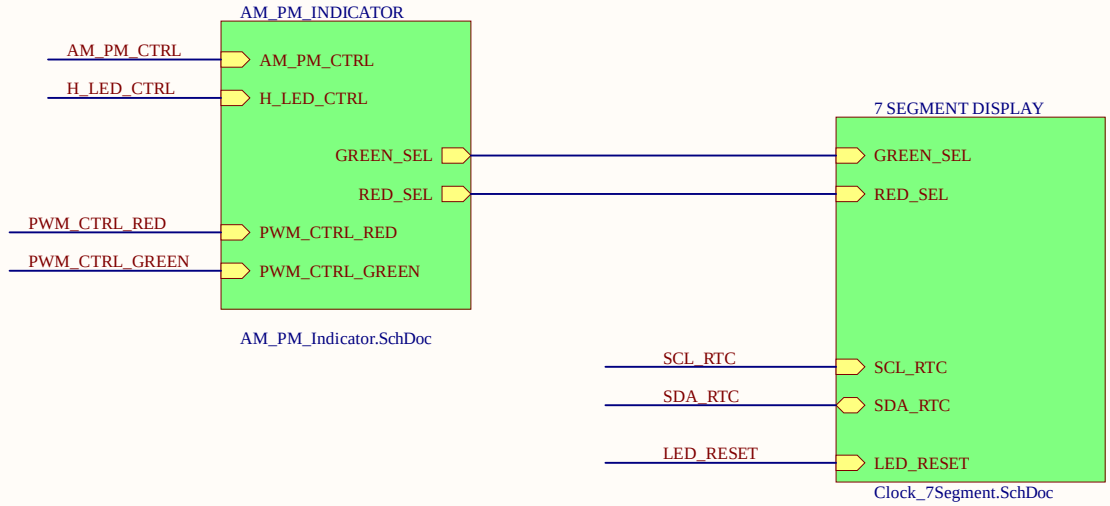
BananaPi-M2 Zero



DBG Connector

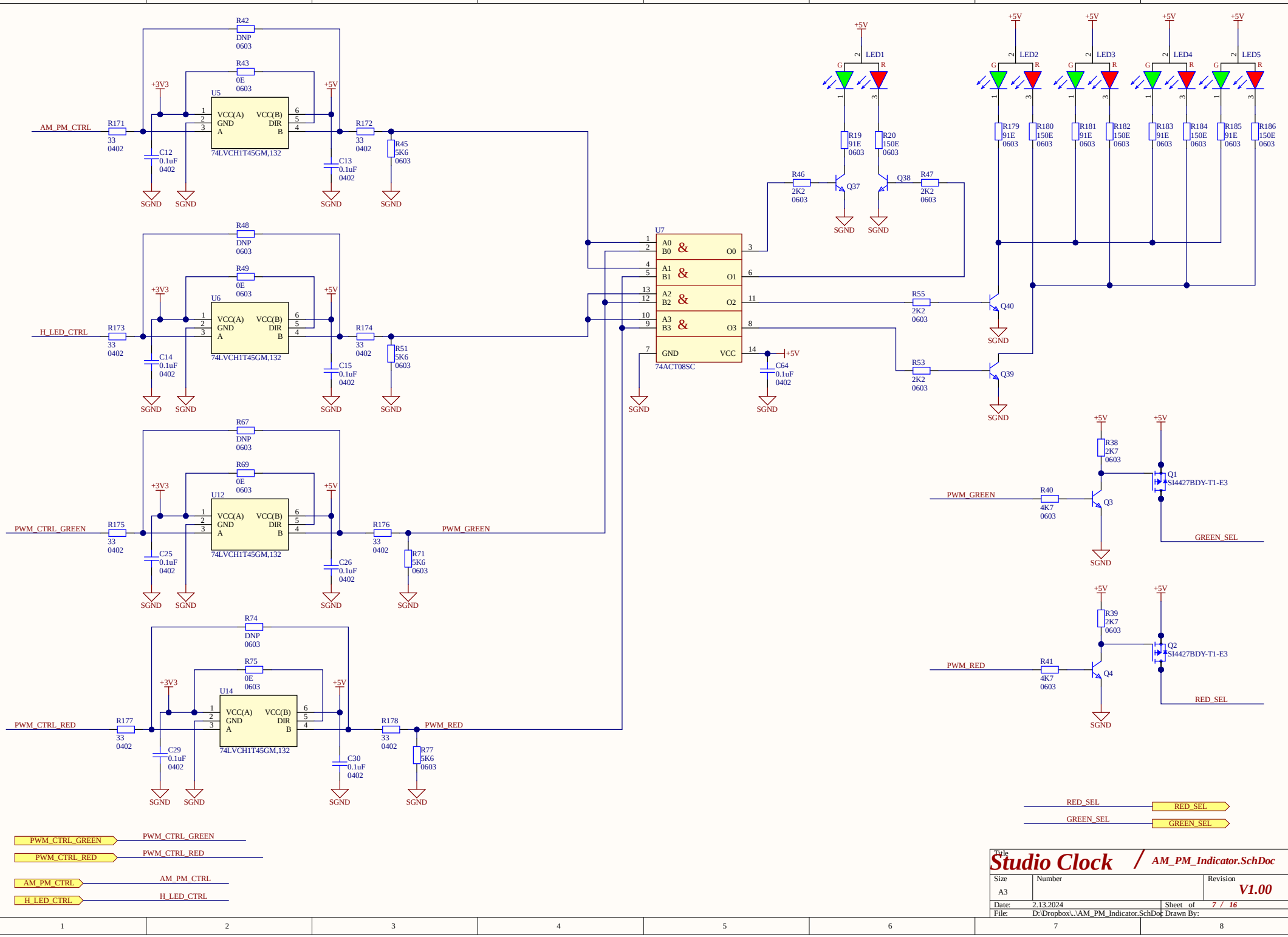


Title: Studio Clock / MP_CTRL1.SchDoc		
Size: A3	Number:	Revision: V1.00
Date: 2.13.2024	Sheet of: 5 / 16	
File: D:\Dropbox\..\MP_CTRL1.SchDoc	Drawn By:	

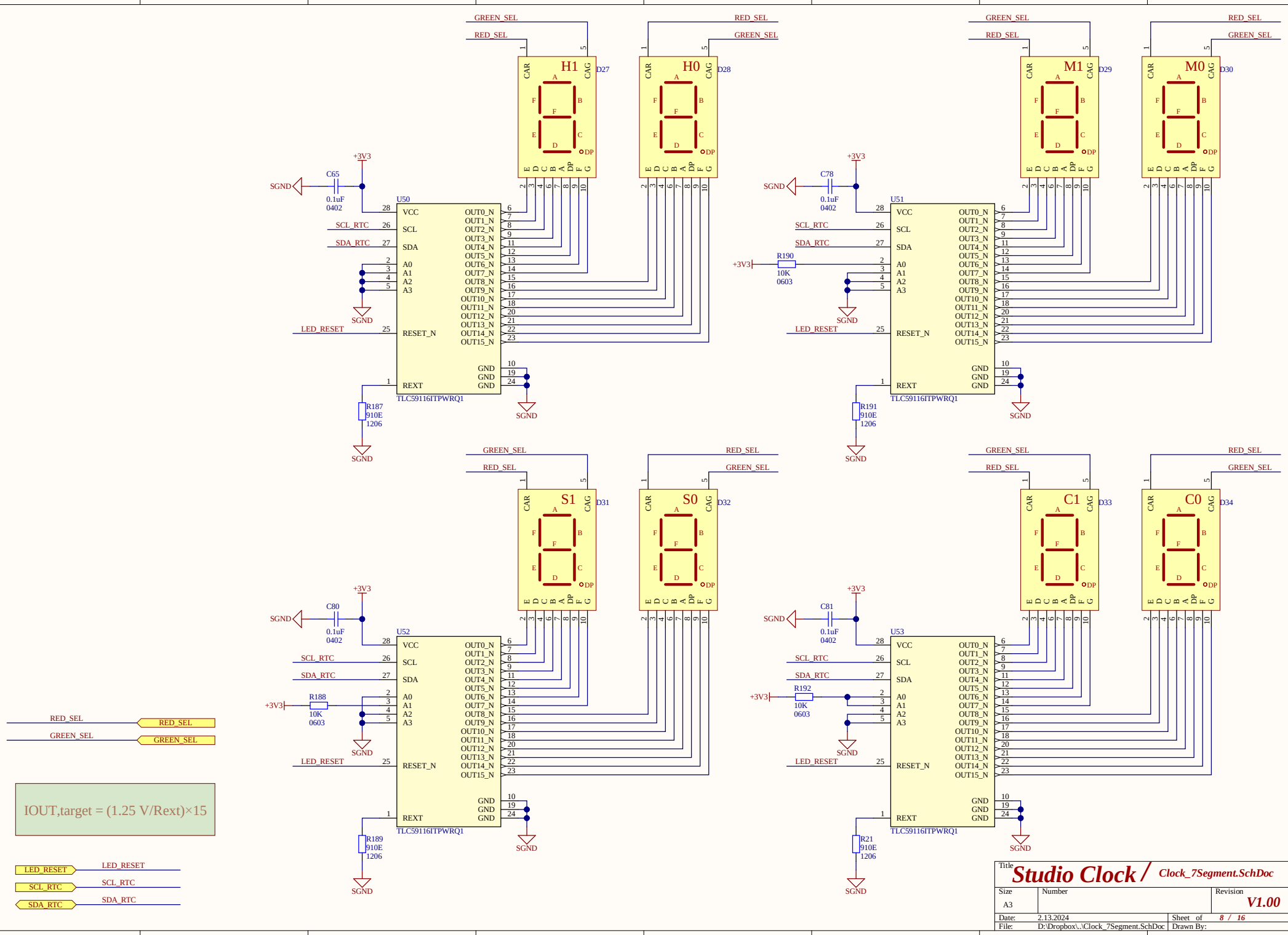


LED_RESET
SCL_RTC
SDA_RTC
PWM_CTRL_GREEN
PWM_CTRL_RED
AM_PM_CTRL
H_LED_CTRL

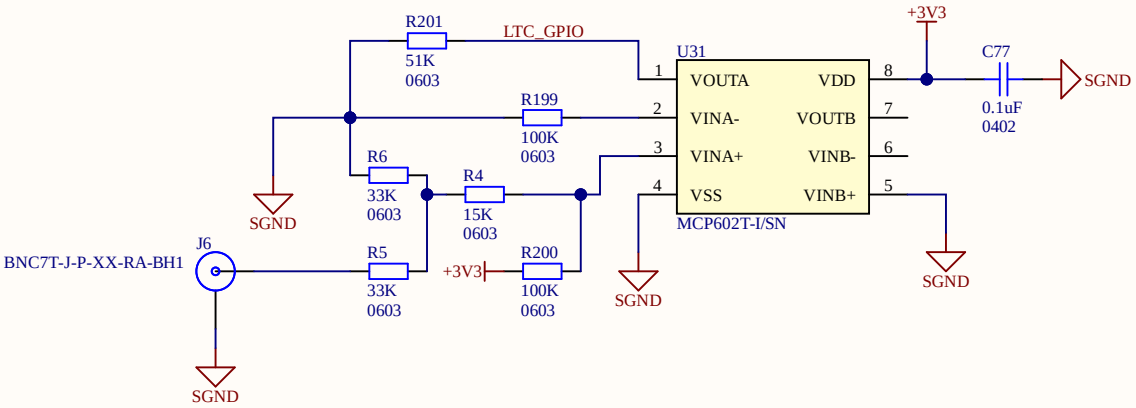
Title: **Studio Clock /** *CLOCK_Display.SchDoc*		
Size: A4	Number:	Revision: ***V1.00***
Date: 2.13.2024		Sheet of ***6 / 16***
File: D:\Dropbox\..\CLOCK_Display.SchDoc		Drawn By:



Title	Studio Clock / AM_PM_Indicator.SchDoc	
Size A3	Number	Revision V1.00
Date: 2.13.2024	Sheet of 7 / 16	
File: D:\Dropbox\..\AM_PM_Indicator.SchDoc	Drawn By:	

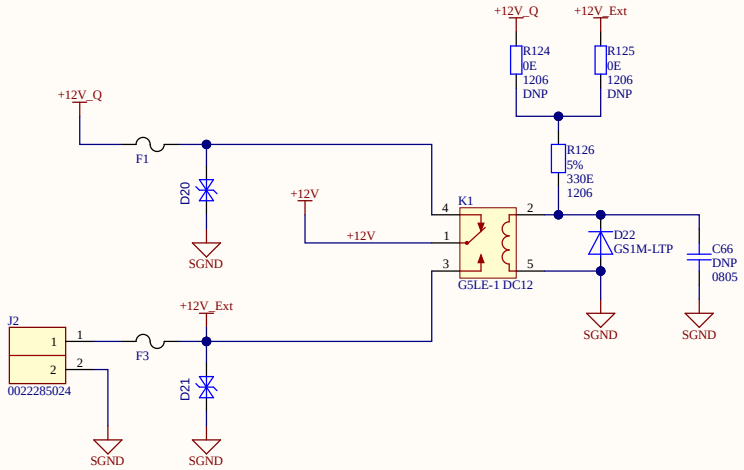


Title	**Studio Clock /** *Clock_7Segment.SchDoc*	
Size A3	Number	Revision ***V1.00***
Date: 2.13.2024	Sheet of ***8 / 16***	
File: D:\Dropbox\..\Clock_7Segment.SchDoc	Drawn By:	

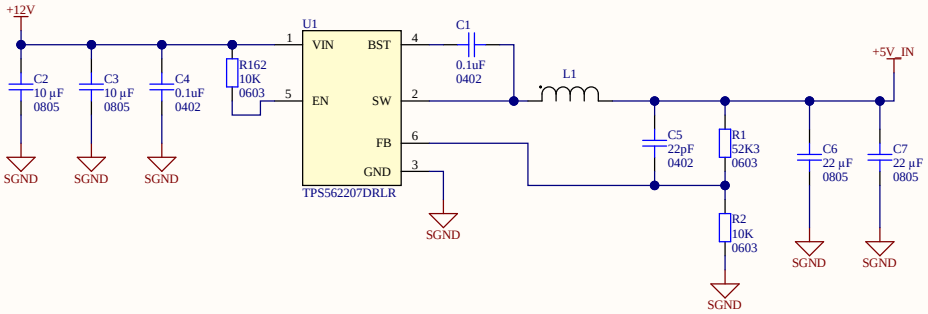


LTC_GPIO LTC_GPIO

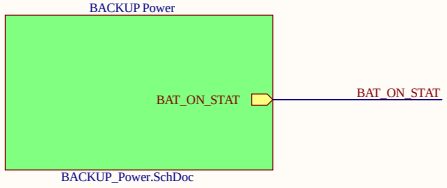
Title	Studio Clock / VIDEO_Interface.SchDoc		
Size A4	Number	Revision V1.00	
Date: 2.13.2024		Sheet of 9 / 16	
File: D:\Dropbox\..\VIDEO_Interface.SchDoc		Drawn By:	

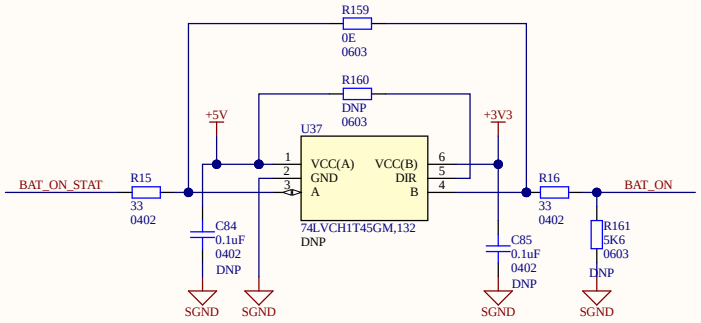


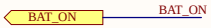
+12V Power selector



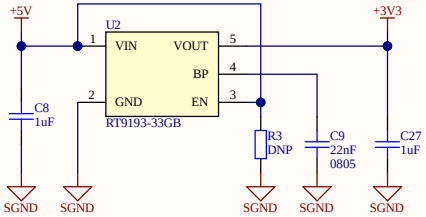
+12V to +5V





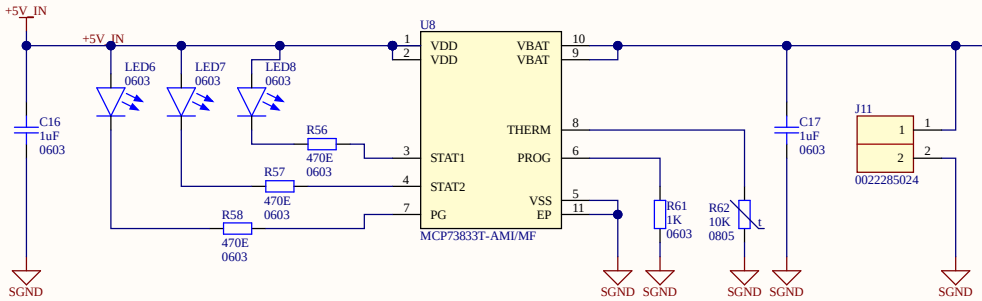


+5V to +3V3

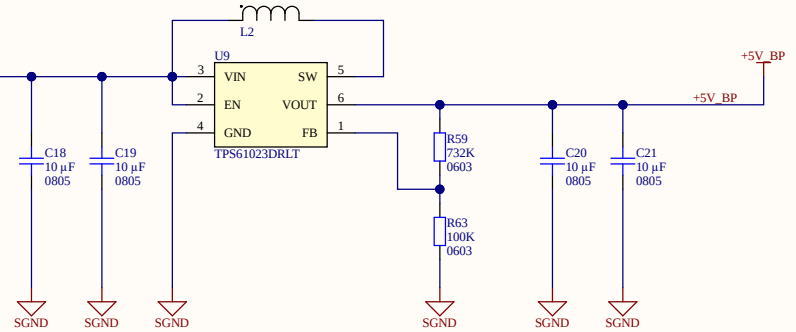


Title: Studio Clock / POWER_Supply.SchDoc		
Size: A3	Number:	Revision: V1.00
Date: 2.13.2024		Sheet of 10/ 16
File: D:\Dropbox\..\POWER_Supply.SchDoc		Drawn By:

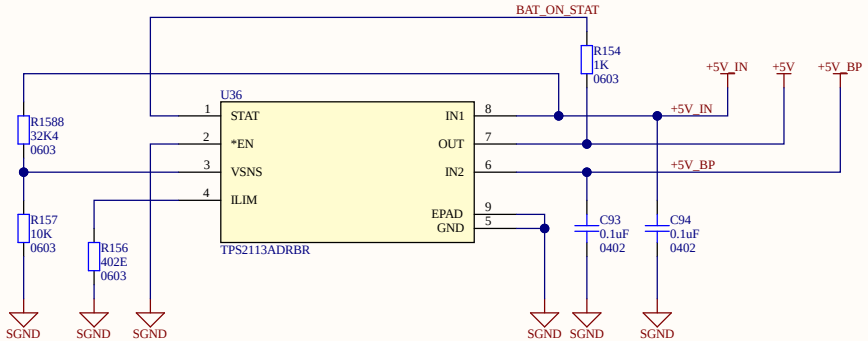
J11 - Battery Connector



Battery Charger



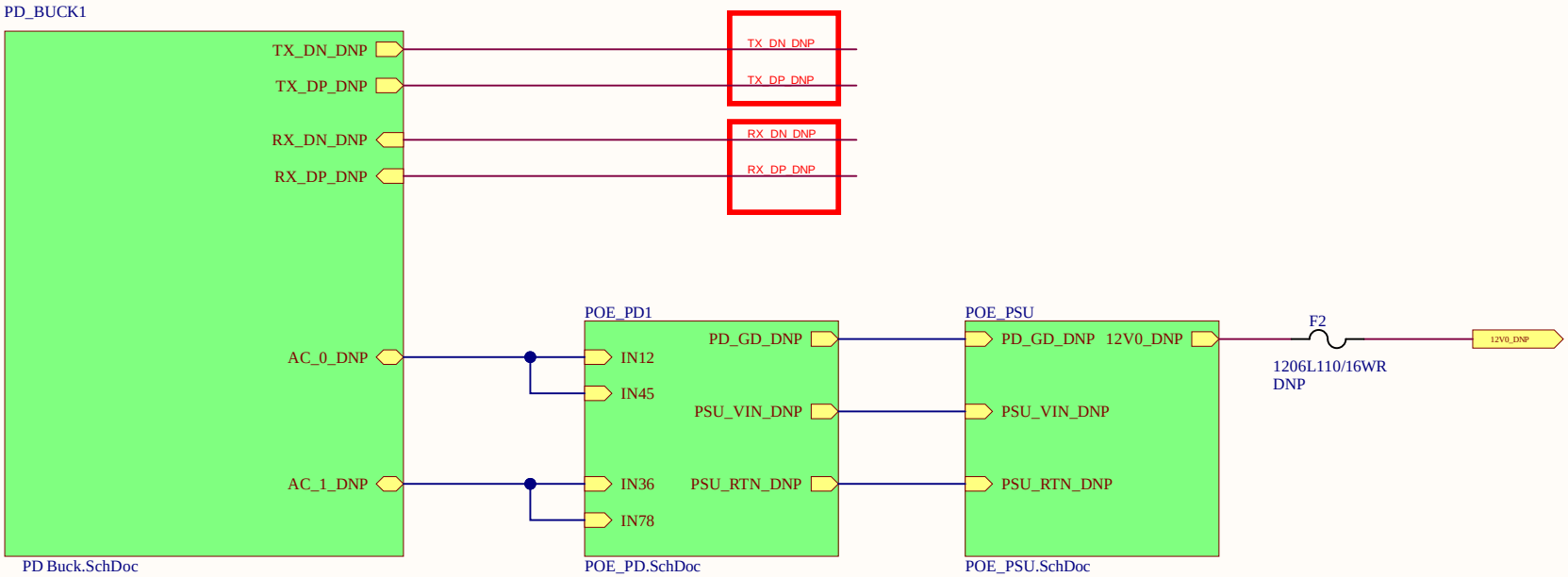
Step-Up (battery - 5V)



POWER MUX

BAT_ON_STAT BAT_ON_STAT

Title: Studio Clock / BACKUP_Power.SchDoc		
Size: A3	Number:	Revision: V1.00
Date: 2.13.2024	Sheet of: 11 / 16	
File: D:\Dropbox\..\BACKUP_Power.SchDoc	Drawn By:	



All POE part was not reviewed and not changed. It was leaved as DNP (Client task).

Title: Studio Clock / POE.SchDoc		
Size: A4	Number:	Revision: V1.00
Date: 2.13.2024	Sheet of 12/ 16	
File: D:\Dropbox\..\POE.SchDoc	Drawn By:	

All POE part was not reviewed and not changed. It was leaved as DNP (Client task).

10/100Mbit RJ45 Jack with integrated Magnetics, CMC, Bob Smith Term and POE.

ESD & Lightening Surge Protection.

CN1
LPJ16801CNL

D1
824014883

DNP

Diff Z = 100 OHM

Diff Z = 100 OHM

DATA OUTPUT

POWER + DATA INPUT

TD+ 1
CT0 2
TD- 3
RD+ 4
CT1 5
RD- 6
PWR+ 7
PWR- 8
SPARE+ 9
SPARE- 10

4 X 0.01uF
100 VDC

75 OHM
75 OHM
75 OHM
75 OHM

1000pF
2KV

MH1 11
MH2 13
MH3 14
MH4 12

R81 75E 0603 DNP
R82 75E 0603 DNP
R83 75E 0603 DNP
R84 75E 0603 DNP

R85 75E 0603 DNP
R86 75E 0603 DNP

U17
2036-07-SMLF

D5
824015031

C34
1n
0603
DNP

DNP

POWER OUTPUT

Cable Screen/RJ45 Jack Shield = Chassis

Lightening Surge Protection Gas Discharge Tubes.

ESD Surge Protection Bidirectional Tranzorb.

C35
1n
1206
DNP

Ethernet Data Centre Tap Termination

R87 0 1210 DNP
R88 0 1210 DNP
R89 0 1210 DNP
R90 0 1210 DNP

PDN_RTN

Connect Shield (Chassis) to 'GND' at one point, closest to either side of the RJ45 connector. Fit 0 Res until EMC/EMI testing is conducted. If it fails fit Ferrite Bead/Cap as a filter to pass the test.

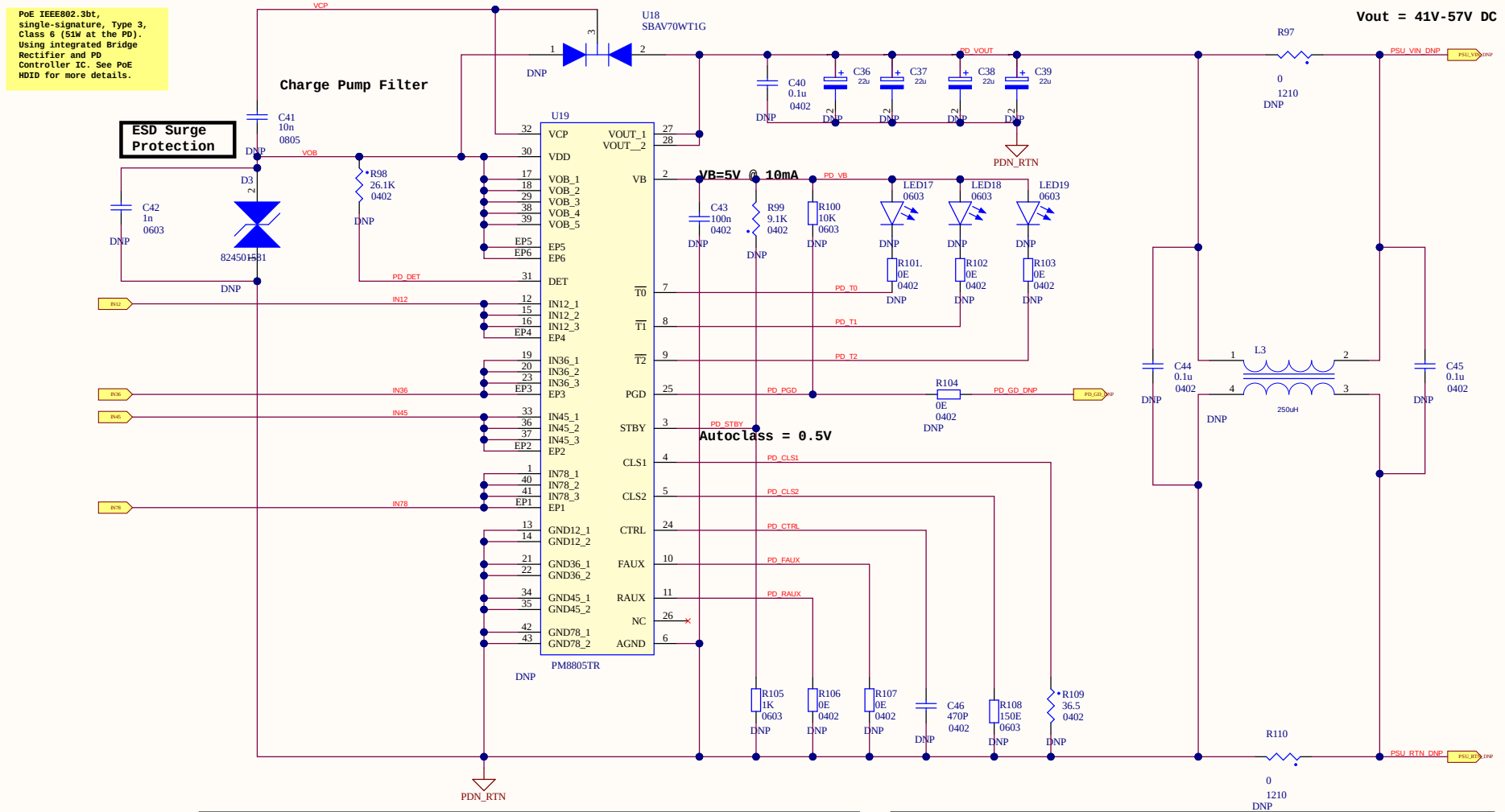
DNP 0 RES to optionally close the Chassis loop around the edge of the PCB

R91
0
0805
DNP

SS1 SPACER
SS2 SPACER
SS3 SPACER
SS4 SPACER
SS5 SPACER

R92 0 1210 DNP
R93 0 1210 DNP
R94 0 1210 DNP
R95 0 1210 DNP
R96 0 1210 DNP

Title: Studio Clock / PD Buck.SchDoc		
Size: A3	Number:	Revision: V1.00
Date: 2.13.2024		Sheet of 13/ 16
File: D:\Dropbox\..\PD Buck.SchDoc		Drawn By:



CLS1 & CLS2: set to Type 3, Class 6 (51W) = T0,T1,T2 = 100.
T0, T1, T2: open-drain o/p, connect RES & LED to VB.
PGD [Power Good]: open-drain o/p. PU to monitor o/p.
CTRL: soft-start cap.
STBY: set to Autoclass. Use Res network to set pin between 0.5V to 1.25V.
FAUX [Front AUX]: not used, PD to 0V.
RAUX [Reart AUX]: not used, PD to 0V.
DET: Signature Detection pin, fit 26K1 to Enable the PSE.

EMI/EMC Common-Mode Filter Circuit.
Caps are always fitted.
Do NOT fit all the parts. Filter Recommended by Silvertel & TI.

Fitting Options:-
1) Default: Fit 0R Res. DNF Common Mode Choke (CMC) L4.
2) Fit L4 CMC & DNF Res R68, R69 if earthing the PSU O/P.
3) DNF CMC & Fit Ferrite Beads FBMH3225HM102NT in place of Res R68, R69 if floating the PSU O/P (ie using 0V GND).

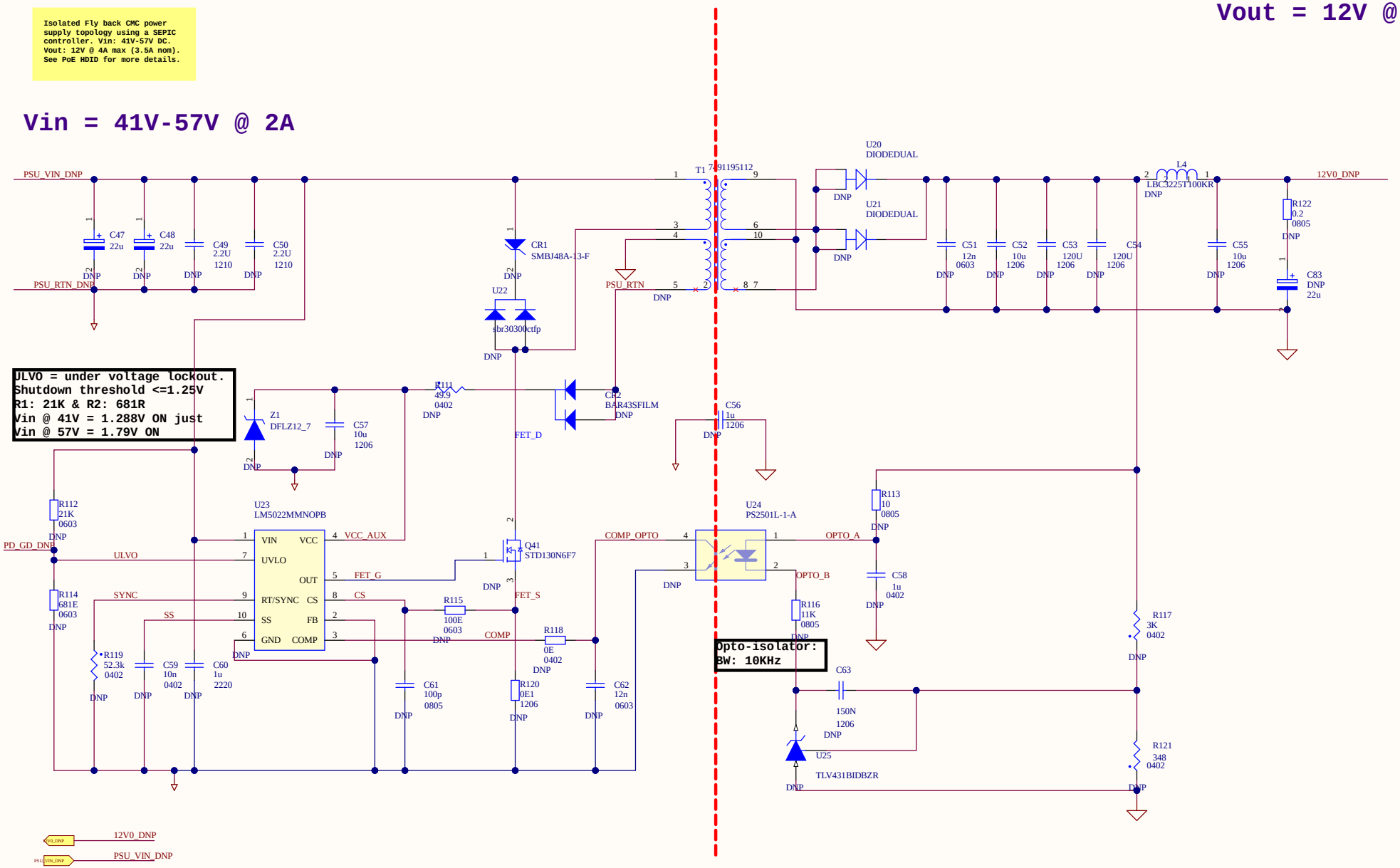
All POE part was not reviewed and not changed. It was leaved as DNP (Client task).

Title	Studio Clock / POE_PD.SchDoc		
Size A3	Number	Revision V1.00	
Date: 2.13.2024		Sheet of 14/ 16	
File: D:\Dropbox\..\POE_PD.SchDoc		Drawn By:	

Isolated Fly back CMC power supply topology using a SEPIC controller. Vin: 41V-57V DC. Vout: 12V @ 4A max (3.5A nom). See PoE HDID for more details.

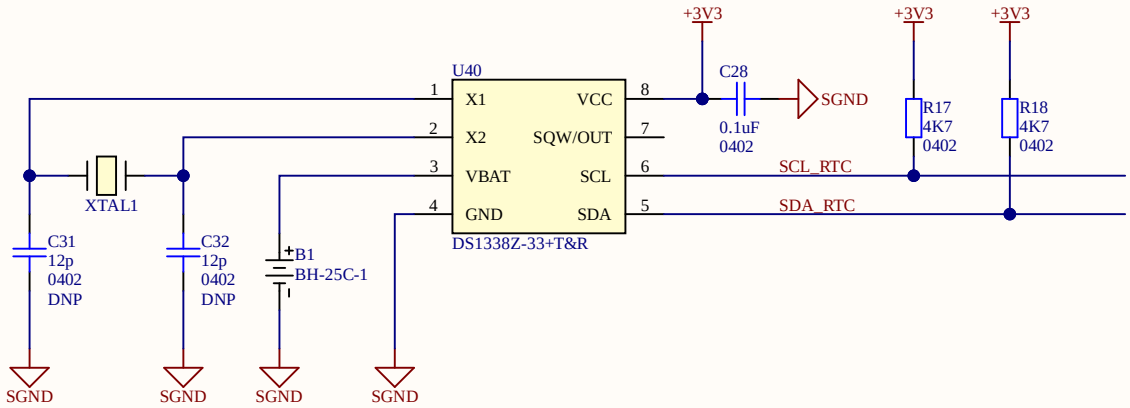
Vout = 12V @ 2A

Vin = 41V-57V @ 2A



All POE part was not reviewed and not changed. It was leaved as DNP (Client task).

Title: Studio Clock / POE_PSU.SchDoc		
Size: A3	Number:	Revision: V1.00
Date: 2.13.2024	Sheet of 15/ 16	
File: D:\Dropbox\..\POE_PSU.SchDoc	Drawn By:	



SCL_RTC SCL_RTC

SDA_RTC SDA_RTC

Title: Studio Clock / RTC.SchDoc		
Size: A4	Number:	Revision: V1.00
Date: 2.13.2024	Sheet of 16/ 16	
File: D:\Dropbox\..\RTC.SchDoc	Drawn By:	